

Handover: the 2026-08-28 arc — the transfer scenario, the shape decision, and a scoping result

Written 2026-08-28. Read `handover.md` first for standing context, then `handover_2026-08-26.md` and `handover_2026-08-27.md` (whose §12 records review gate R.1). This file covers only this arc. It is written by the session that did the work and inherits that session's blind spots; where it disagrees with the repository, the repository wins.

0 What this arc was

It began as card B.1 — build the cyclist-overtake field — and turned into a scoping result about what the study-1 data can and cannot establish. Nothing that was previously computed has been found *wrong*; what changed is what those computations can be claimed to show. Two datasets arrived mid-arc (`02_Cut-in`, `03_CAMP`) and the first of them answered a question that had been referred to a review gate.

1 The finding that reframes the rest

In the study-1 cut-in design, relative speed is constant at 2.78 m/s across every one of the 18 cells, so $\text{correlation}(\text{gap}, \text{TTC}) = 1.0000$. Time headway and required deceleration are fixed functions of the same single number. The longitudinal dimension of that design has one degree of freedom.

Everything longitudinal fitted on study 1 — the stage-0 correlation of 0.90, the stage-1 boundary level of 5 352, the population percentile — is therefore equally consistent with a driver who thresholds the preference field and a driver who thresholds **the gap**. The fits are not wrong and the boundary is well estimated on its own scale. But they cannot be cited as evidence that the field's kinematic content is the right description, because the design contains no contrast that separates it from the simplest possible alternative. Reproduced by `replication/czb/cutin2_scope.py` §0.

This should have been noticed earlier. It was not, and the practical cost is that several sessions of work were spent refining an estimator on a design that cannot discriminate the model it estimates.

2 The second cut-in study breaks the collinearity, and rules against us

`02_Cut-in`: 10 944 trials, 168 participants, the same three response variables as study 1 including the ordered braking question. Its delta-velocity factor is crossed with TTC and the two multiply into distance by construction, so matched-TTC cells span gaps from 1.6 m to 78 m — six-fold, in the scenario whose field already exists.

- **All 24 matched-TTC rows run negative.** At identical time-to-collision, a larger gap means less intervention, by as much as 0.042 against 0.818 at TTC 4.7 s. Time alone does not determine the response.
- Over all cells: **gap orders the response at $p = -0.887$** , time-to-collision at **-0.807** , and required deceleration at **$+0.238$** . The last has the sign a demand-based model wants but a negligible magnitude — so the quantity our safety terms are built from barely orders these cells while gap orders them almost perfectly.
- It cannot separate gap from closing speed or headway, which are perfectly confounded at matched TTC by construction. It shows a time measure is insufficient without saying which ingredient is missing.

`03_CAMP` is described only (19 files); no use proposed, per Jonas.

3 Card B.1: the field does not represent the cyclist overtake

`src/comfortzone/overtake.py`, `docs/overtake_construction_note.md`. The loader is validated against the study's own labels (edge clearance at the pass 0.506 / 1.004 / 1.501 m against 0.5 / 1 / 1.5). Three corrections to the card: roles must come from the instructed vehicle, because the cut-in's lateral-span rule is **inverted** here (the ego moves 2.24 m, the cyclist 0.16); the lateral coordinate is `Location_Y`, since `Offset` inverts the criticality ordering; and the cyclist's width is in the trace (0.582 m).

The card's premise fails. The field orders the 15 cells at Spearman **$+0.402$** while the clearance label alone orders them at **-0.833** , because `p_lane` sits at 0.843–1.000 through the entire response window and cannot see the manipulated variable. The scenario's criticality is a lateral **comfort margin** — 0.5 m is uncomfortable while being uncontroversially collision-free — and the preference function's lateral machinery is collision-oriented. A symmetric-projection repair (`lane_entry_bidirectional`) was built, tested, and **over-corrects**: `p_lane` collapses to

0 by C3 and the cells end up ordered -0.537 . Kept, default off, as tested-and-rejected.

But the transfer itself works once the baseline is matched. Frozen transfer scores RMSE 0.176 against chance 0.158 — worse than chance — while freeing **only the per-scenario lapse** gives 0.129, past chance and within 0.016 of the full-refit ceiling 0.113. The pre-onset rate is 0.081 in the cut-in and 0.198 in the overtake, so a no-shift primary is mis-specified at the baseline before the boundary is consulted. The lapse is identified by C1 cells alone. Roadmap §2 step 4 and card B.4 amended accordingly.

4 The one-scalar claim has independent support

replication/czb/cross_scenario_consistency.py, and this needs no field at all. All 43 participants saw all four Random scenarios, so the claim's central prediction — that a driver carries one level — is testable directly. Per-driver criticality-adjusted propensity correlates **+0.50 to +0.74** across the six scenario pairs, which is **0.53–0.78 of the split-half reliability ceiling** (reliabilities 0.93–0.98), mean **0.69**.

About 69% of the reliable per-driver signal is shared across scenarios. That is the strongest evidence for the CZB framing that does not depend on any field construction, and it simultaneously caps stage B: a third of reliable per-driver variance is scenario-specific, so B.4 should be scored against ~ 0.69 rather than against perfection.

5 The S-shaped lane-entry ramp, decided

Jonas's proposal, implemented as `lane_entry_shape(u, k)` — a normalized logistic with $g(0) = 0$, $g(1) = 1$ exactly for every k and $g(u; 0) = u$, so the linear ramp is nested. Swept on the cut-in surface with the stage-0 probit refitted at each k : RMSE 0.1189 linear, minimum **0.1154 at $k \approx 12$** , reflected shape worse at 0.1211. Direction supported, gain modest (2.9%). The $k \rightarrow \infty$ limit is the released binary gate and is the worst of the family (0.1457), which independently corroborates the 2026-08-27 continuity decision.

Decided by Jonas: k is fixed at 12, applied on the CZB staging path only

(`comfortzone.cutin.CZB_LANE_ENTRY_SHAPE_K`); PreferenceParams still defaults to 0 so causation work and released comparisons are untouched. Documented as a *calibrated constant, not parameter-free* — chosen once against the cut-in response surface, same standing as the released model's per-scenario $a_{OV,min}$ — with the rule that it must never become a per-scenario knob. **Consequence not to lose:** it changes the covariate in 3 of 18 cut-in cells (TTC4/C2 +26%, TTC6/C1 -27%, TTC8/C1 +8%), which are the lapse-identifying cells, so stage 1 and the percentile table must be regenerated before being quoted (card A.2.v2). **A first measurement of that consequence, seen in passing when the stage-2 refit re-fitted stage 1 under the new constant: the between-driver spread moves from 0.209 to 0.194**, about 7%. So every percentile quoted in this file and in `handover.md` is a **pre- $k=12$ number** and will shift by roughly that much. They are kept as written because they are what the committed outputs currently say; A.2.v2 replaces them.

6 R.1.Q2 and R.1.Q3, and a defect they exposed

The fitted lapse–threshold correlation was -0.727 under an independence assumption. Simulating **independent** driver effects and refitting recovers -0.272 , so 37% is manufactured by the estimator's lapse/threshold trade-off; but the observed value is 2.9 sd below that arm and inside the arm simulated with a genuine -0.7 , so some is real. A.2.v2 must judge its fitted correlation against -0.27 , not zero.

Two discriminating tests were then run. **The floor is a person trait:** per-driver C1 rate correlates +0.546 between cut-in and cyclist overtake, 61% of the reliability ceiling. **Whether it is learned during the study is untestable on study 1:** the within-session slope is +0.005 (SE 0.052), but a participant presses at C1 a median of *once* in 24 trials, 17 of 43 drop out for having no variation and those are the low-floor participants, and the smallest detectable slope is 0.146 against a real effect of +0.027. That is not evidence of absence. The second cut-in study, which shows every block-1 clip twice, gives the answer: $P(\text{intervene})$ rises from **0.547 to 0.575** between first and second showing of an identical clip (3 456 pairs, SE 0.007). Jonas's anticipation account is supported.

And C1 is not a null scene. At C1 the three cut-in conditions differ by a factor of two in car-following state (gap 10.5 / 16.0 / 21.5 m, THW 0.34 / 0.52 / 0.70 s) and behaviour is ordered accordingly (0.122 / 0.070 / 0.052) — an ordinary response to an ordinary car-following situation, not anticipation. **The field ranks those cells backwards**, assigning deficit 1 / 1509 / 2907. So the lapse is being fitted to cells carrying real boundary signal with a covariate that ranks them wrongly: a fifth explanation for R.1.Q2 that was on nobody's list, and the only one that is a defect in our own construction. Blocker B2.Q1.

7 What changed on disk

New: src/comfortzone/overtake.py; docs/overtake_construction_note.md, docs/lateral_and_uncertainty_note.md, this file (each with Word/PDF); replication/czb/{overtake_field_check, transfer_cutin_to_overtake, cross_scenario_consistency, lane_entry_shape_check, lapse_threshold_artifact, response_style_and_anticipation, cutin2_scope}.py with tracked outputs and logs. Modified: src/aidriver/preferences.py (lane_entry_shape, lane_entry_shape_k, lane_entry_bidirectional, lane_entry_max_dy_m — all defaulting to released behaviour), src/comfortzone/cutin.py (the k constant and staging), src/comfortzone/czb_data.py (overtake trials, params pass-through), tests/test_cutin.py (19 → 73 checks), the roadmap, work orders, README, and handover.md. Suite 123 → 177, green throughout.

8 Where I would pick up

- 1 **Card B.2.v2's first item — the C1 defect.** It is a blocker on the *interpretation* of every fit so far, and it is cheap to diagnose.
- 2 **Review gate R.2** (below), which now has a much better dataset than when it was written.
- 3 **Card A.2.v2**, which must run before any percentile is quoted, because of $k = 12$.
- 4 Then B.4, scored against the 0.69 ceiling, with the per-scenario lapse free.

7b The $k = 12$ lock has already moved the stage-2 numbers

Recorded because it is the first concrete measurement of the regeneration debt and it arrived after the rest of this file was written. Re-running card A.3 with the constant in place refits stage 1, and the results move:

quantity	$k = 0$ (as first reported)	$k = 12$ (current committed output)
between-driver spread sigma_pop	0.209	0.194
accumulator, in sample	0.178	0.169
accumulator, leave-one-criticality-out	0.267	0.255
stage-1 static threshold, same folds	0.161	0.169
stage-0 static probit, same cells	0.125	0.115

The A.3 verdict is unchanged — held-out 0.255 against a rule of $\leq 0.11 / > 0.13$ is still FAIL, and the accumulator still fits worse in sample than the static probit it extends. So none of this arc's conclusions move. What it does show is the size of the regeneration debt: everything lapse- or percentile-related that predates 2026-08-28 carries a shift of this order, and card A.2.v2 now states the expected magnitude so an anomalous regeneration is detectable.

8b Known gaps this arc did not close

Listed so they are not rediscovered as surprises.

- **The handbook contains none of this arc.** Chapters 04, 08 and 11 still describe the field, the accumulator and the CZB path as they stood on 2026-08-27. Jonas's Word review is still pending, so no $\{\{R5\}\}$ round was started; but a reader of the handbook today would not learn that the accumulator failed, that the overtake field does not represent its scenario, or that study 1's longitudinal design is degenerate.
- **Card C (percentile sensitivity) has never been run.** It now depends on A.2.v2, since $k = 12$ changes the percentile table.
- **No per-criticality deficit figure for the overtake.** Card B.1 asked for one; the numbers are in out/overtake_field_check.md but the figure was not produced.
- **The query identifiers are inconsistently formed:** this arc introduced B2.Qn alongside the earlier B.1.Qn and R.1.Qn. They denote different cards and the collector parses both, but the reader should not assume B2 and B.1 are related. Identifiers are never rewritten once raised, so this stands as a wart.
- **02_Cut-in has not been fitted**, only scoped. Its traces do load with the existing cut-in loader (checked: ego 28.7 m/s, target width 1.88 m, onset detected), so review gate R.2's first task is not blocked on pipeline work.

9 Caveats on this file

The transfer result in §3 rests on 15 cells with a narrow response range (0.14–0.69) and one condition that is flat; RMSE differences of a few hundredths there are not decisive. The 69% in §4 is a correlation of rate residuals, not of

fitted boundary levels — it supports the trait claim, it does not establish that the trait is *the comfort-zone level*. And §1's collinearity finding is about the longitudinal dimension only; the lateral manipulations in study 1 are genuinely independent of it.